Thm. Let R be an integral domain. Then, the Evaluate map

$$\Phi : R[x] \to R^R : f \mapsto (a \mapsto f(a))$$

is injective.

Proof. Notice that: $R^X := \{f : X \to R\}$ is a Ring, with

$(f+g)(x) := f(x) + g(x)$ and $(fg)(x) = f(x) \cdot g(x)$.

And, Φ is a Ring-Homomorphism, because: for any $f, g \in R[x]$, $a \in R$,

$\Phi(f+g)(a) = f(a) + g(a) = \Phi(f)(a) + \Phi(g)(a) = (\Phi(f) + \Phi(g))(a)$

$\Phi(fg)$